LECTURE 15
Burgers Equation: Shocks, Rarefaction and Conservation Form

Reference: J. C. Tannehill, D. A. Anderson & R. H. Pletcher, Computational Fluid Mechanics and Heat Transfer, 2nd ed., Taylor & Francis, 1997. Sections 3.3.7, 4.4 and 4.5.
L. A. Barba & G. F. Forsyth, 12 steps to Navier-Stokes, step 4.
Wladimir E. Banda-Barragán

Burgers Equation

Burgers (1948) introduced the simplest equation that carries the three terms of the fluid equations: **an unsteady term, a nonlinear convective term and a diffusive term.**

$$\underbrace{\frac{\partial u}{\partial t}}_{\text{unsteady}} + \underbrace{u\frac{\partial u}{\partial x}}_{\text{convective}} = \underbrace{\mu\frac{\partial^2 u}{\partial x^2}}_{\text{viscous}}$$

With the viscous term, the equation is **parabolic**.

Dropping it leaves a **hyperbolic** equation (non-linear advection):

$$\boxed{\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = 0}$$

Each point of the profile travels at its own value of u, so the wave speed changes across the wave and a smooth profile can steepen into a discontinuity.

Conservation Form

Divergence form:

every spatial derivative appears inside one divergence, so the equation states that a quantity is conserved:

$$\frac{\partial u}{\partial t} + \frac{\partial F}{\partial x} = 0, \qquad F = \frac{u^2}{2}$$

Non-conservative, or non-divergence, form:

This is the same equation with the flux differentiated by the chain rule:

$$\frac{\partial u}{\partial t} + A\,\frac{\partial u}{\partial x} = 0, \qquad A = \frac{dF}{du} = u$$

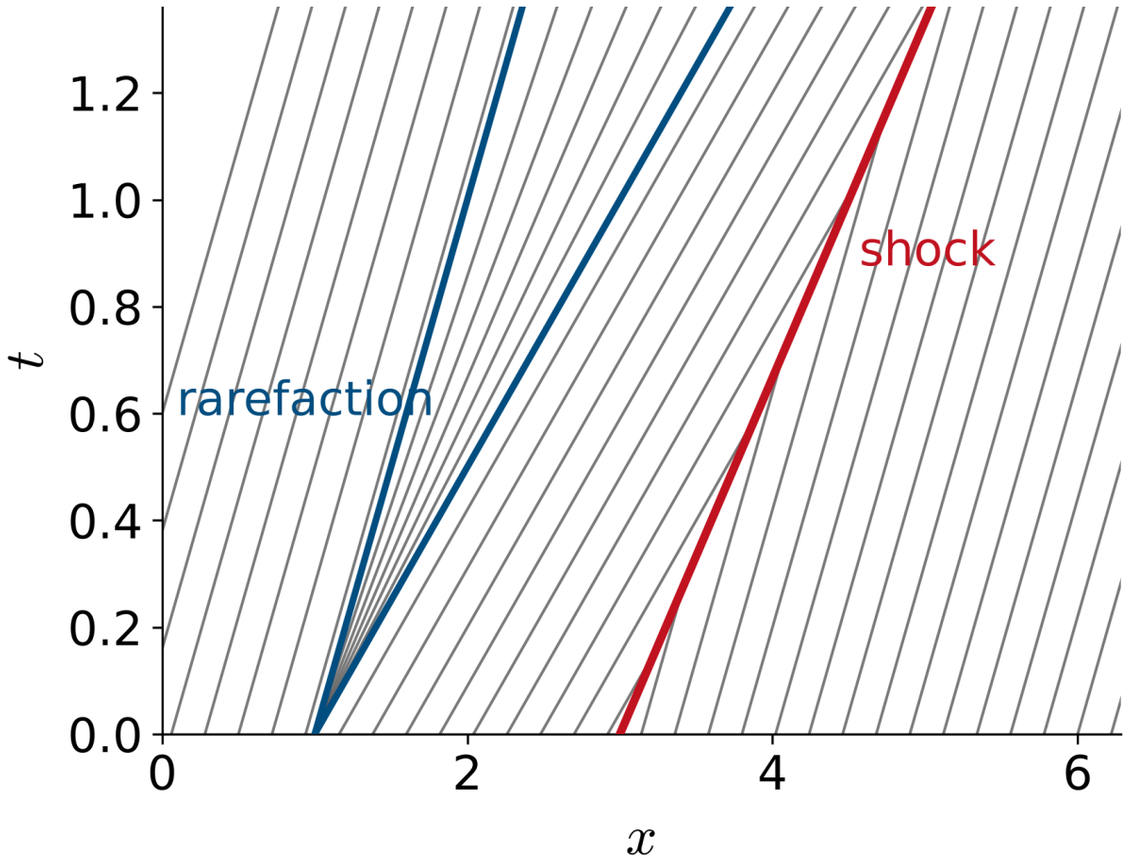
The two are identical wherever u is differentiable.

They are not identical where u jumps, and that is the case we study here.

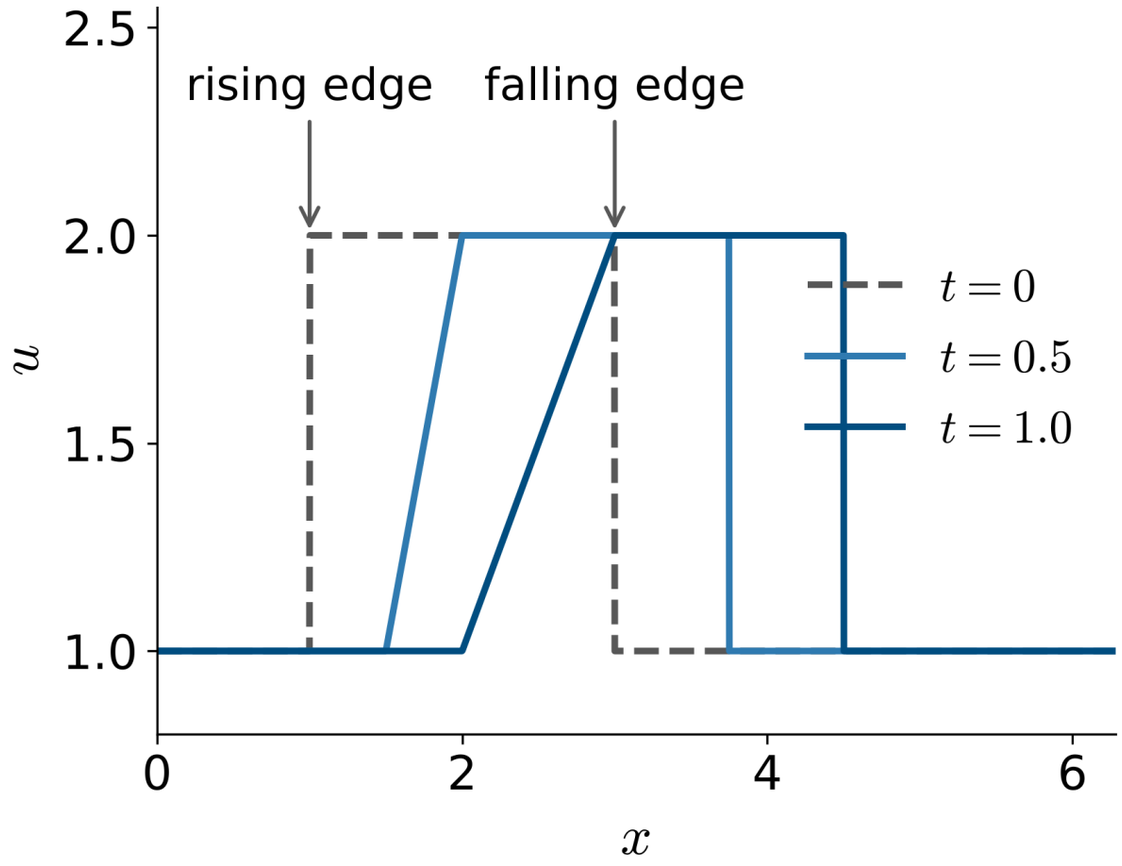
Characteristics: Rarefaction and Shock Formation

We model a top-hat (square) pulse of $u = 2$ on a background $u = 1$.

Characteristics



Effects on rising and falling edges



Two effects can be seen:

- The rising edge at $x = 1$ opens into the **rarefaction fan**.
- The falling edge at $x = 3$ collapses into the **shock**.

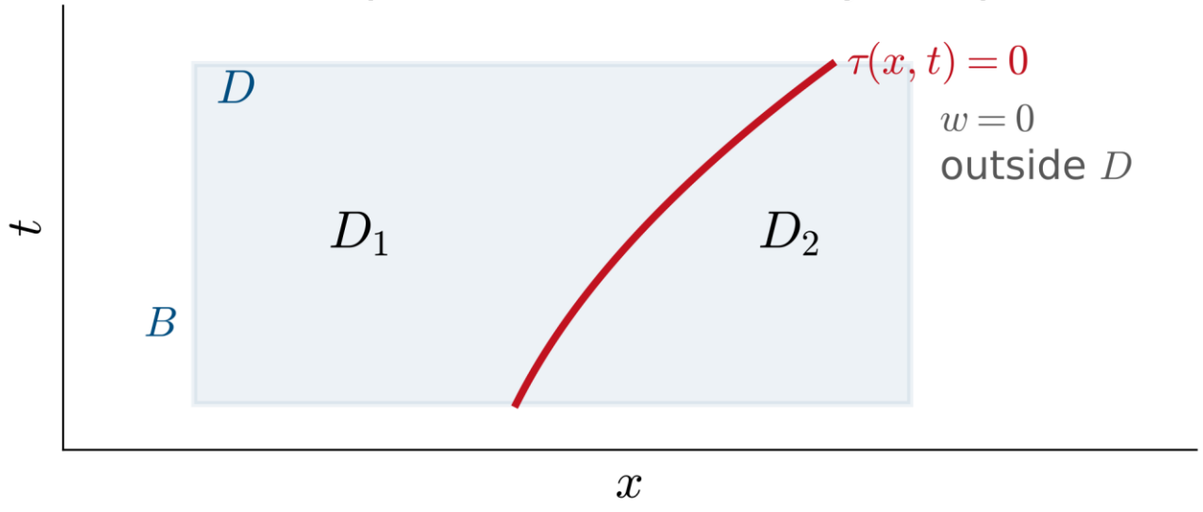
Weak Solutions

Once u is discontinuous the derivatives in the equation do not exist. Lax (1954) replaces them by an integral statement over a rectangle D of the (x,t) plane, for every test function w:

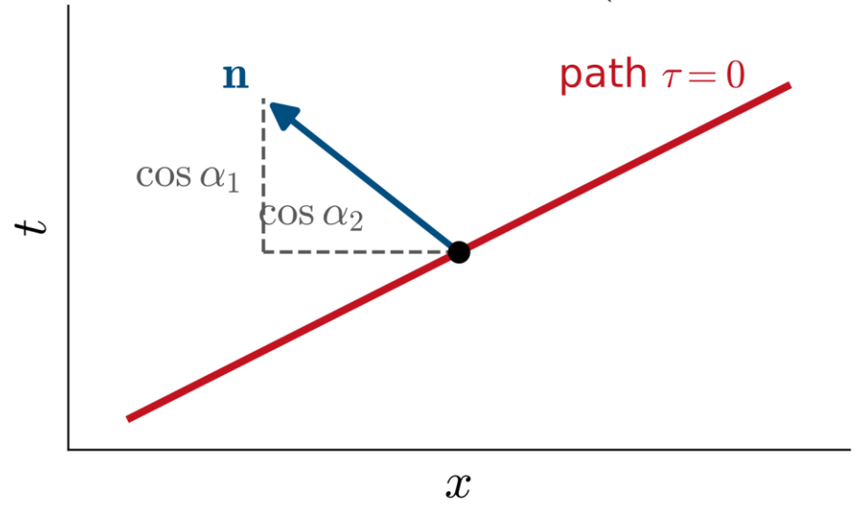
$$\iint_D (w_t\, u + w_x\, F)\, dx\, dt + \int w(x,0)\, \phi(x)\, dx = 0, \qquad \phi(x) = u(x,0)$$

D is any rectangle in the (x,t) plane; w is smooth and vanishes on the boundary B of D and outside it. The second integral carries the initial data.

(a) D, split into D_1 and D_2 by the path



(b) the unit normal, $\mathbf{n} = (\cos\alpha_2,\ \cos\alpha_1)$



Genuine and weak solutions:

- A genuine solution is continuous and has derivatives.
- A weak solution satisfies the integral statement and may jump. Every genuine solution is weak; the converse is not true.

The Jump Condition

We integrate the weak form by parts over D_1 and D_2 separately. The integration by parts leaves a contribution on the path itself:

$$\iint_D (w_t\, u + w_x\, F)\, dx\, dt + \int w(x,0)\, \phi(x)\, dx = 0, \qquad \phi(x) = u(x,0)$$

$$\underbrace{\iint_{D_1} (u_t + F_x)\, w\, dx\, dt}_{=\ 0} + \underbrace{\iint_{D_2} (u_t + F_x)\, w\, dx\, dt}_{=\ 0}$$

Only the line integral along the path survives, and it must vanish for every w:

The two area integrals vanish:

u is smooth inside D_1 and inside D_2 and satisfies the equation there.

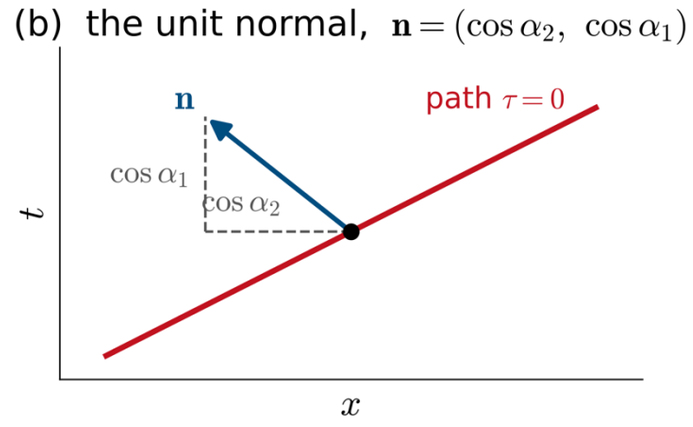
Only the line integral along the path survives, and it must vanish for every w:

$$[u]\cos\alpha_1 + [F]\cos\alpha_2 = 0, \qquad [\,\cdot\,] = \text{jump across the path}$$

This is the Rankine-Hugoniot condition:

The jump sweeps up u at the rate $s[u]$ and the fluxes deliver it at the rate $[F]$, so it is conservation applied across the discontinuity.

Shock Speed



The direction cosines are the components of the unit normal to the path. Writing the path as $t = t(x)$ and putting them in:

$$\cos\alpha_1 = \frac{1}{\sqrt{1+t'^2}}, \qquad \cos\alpha_2 = \frac{-t'}{\sqrt{1+t'^2}}, \qquad t' = \frac{dt}{dx}$$

$$\Longrightarrow \qquad s\,[u] = [F]\,, \qquad s = \frac{dx}{dt} = \frac{F_2 - F_1}{u_2 - u_1}$$

For Burgers' equation:

Substituting $F = u^2/2$ into the **Rankine-Hugoniot condition**:

$$s = \frac{u_2^2/2 - u_1^2/2}{u_2 - u_1} = \frac{(u_2 - u_1)(u_2 + u_1)}{2(u_2 - u_1)} \qquad \Longrightarrow \qquad \boxed{\frac{dx}{dt} = \frac{u_1 + u_2}{2}}$$

The discontinuity travels at the average of the two states, not at the speed of either one.

With $u_1 = 2$ and $u_2 = 1$ it moves at 1.5, while the characteristics behind move at 2 and those ahead at 1, so they run into it from both sides.

Rarefaction

Where the state on the left is the slower one the characteristics spread apart and leave a wedge of the (x, t) plane empty. The characteristic condition is:

$$\frac{dt}{dx} = \frac{1}{u}$$

Filling the wedge with a fan from the origin:

for a jump from $u = 0$ to $u = 1$ the solution is continuous for every $t > 0$:

$$u = 0 \quad (x \leq 0), \qquad u = \frac{x}{t} \quad (0 < x < t), \qquad u = 1 \quad (x \geq t)$$

The width of the expansion grows linearly with time.

Shocks and rarefactions are the two kinds of structure the inviscid equation produces from discontinuous data.

Conservative Property

A scheme written as a difference of fluxes between cell faces inherits the conservation statement of the equation:

$$u_i^{n+1} = u_i^n - \frac{\Delta t}{\Delta x}\left(F_{i+1/2}^n - F_{i-1/2}^n\right)$$

Summing over the whole mesh:

Every interior flux appears twice with opposite signs and cancels exactly, so only the two boundary fluxes are left:

$$\Delta x \sum_{i=1}^{N} \left(u_i^{n+1} - u_i^n\right) = -\Delta t \left(F_{N+1/2}^n - F_{1/2}^n\right)$$

The cancellation is exact at any mesh size, not only in the limit.

A scheme with this property is said to have the **conservative property**, and it is **what forces the computed discontinuity onto the Rankine-Hugoniot speed**.

Non-Conservative Form: Wrong Shock Speed

Written in conservation form:

Upwind differencing of the flux, valid for $u > 0$ is:

$$u_i^{n+1} = u_i^n - \frac{\Delta t}{\Delta x}\left(F_i^n - F_{i-1}^n\right), \qquad F = \frac{u^2}{2}$$

Written in non-conservative form:

The same differencing applied to the chain-rule version:

$$u_i^{n+1} = u_i^n - u_i^n \frac{\Delta t}{\Delta x}\left(u_i^n - u_{i-1}^n\right)$$

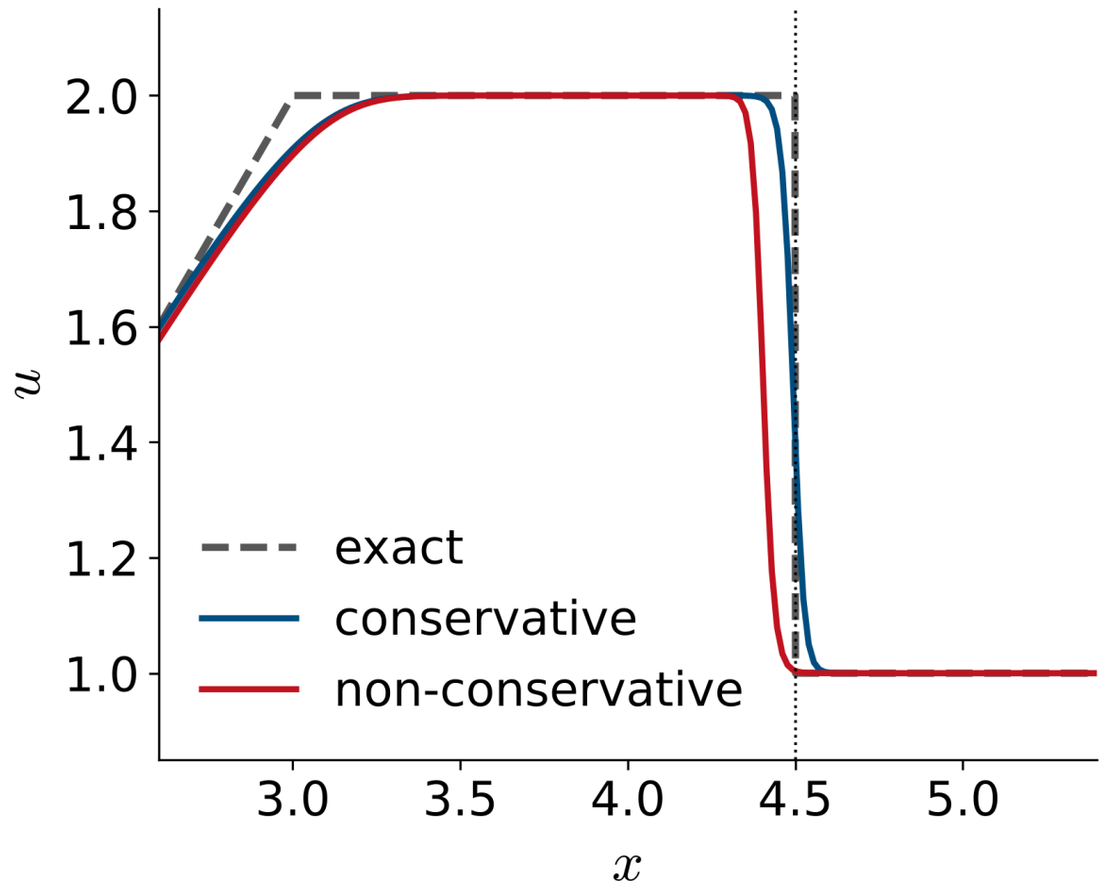
The second has no flux difference to cancel, so nothing telescopes and the discrete total of u is not conserved.

On smooth data both converge to the same answer. Across a shock the second converges to the wrong speed, and refining the mesh does not fix it.

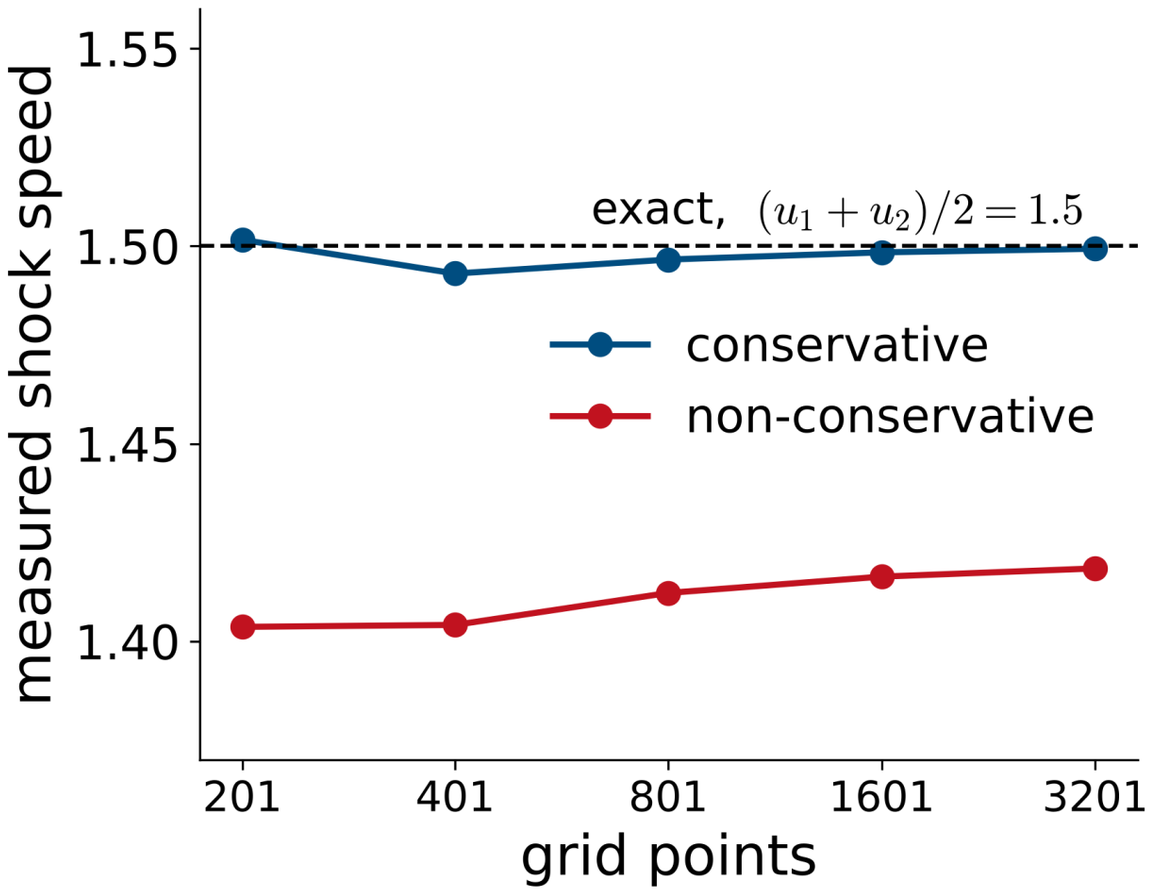
Computed Shock Speeds

The square pulse at $t = 1$, Courant number 0.4.

Shock speed comparison



Shock speed for different resolutions



The non-conservative form converges to 1.42, not to 1.5, regardless of Δx .

See: https://github.com/wbandabarragan/computational-fluid-dynamics/blob/main/notebooks/nb09_burgers_shocks_conservation.ipynb

The Viscous Burgers Equation

Putting the viscous term back makes the equation parabolic.

It is then a model for the **boundary-layer equations** and for the complete **Navier-Stokes equations**:

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = \mu \frac{\partial^2 u}{\partial x^2}$$

Linearised:

Replacing u in the convective term by a constant c gives the linear case with diffusion:

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = \mu \frac{\partial^2 u}{\partial x^2}$$

Notice that:

- At $\mu = 0$ this is the wave equation (Lecture 13).
- At $c = 0$ it is the heat equation (Lecture 12).

The Stationary Viscous Shock

Rakich (1978) combined the two into one equation with free parameters b and c:

$$u_t + (c + b\,u)\,u_x = \mu\,u_{xx}$$

For one choice of the parameters, the equation has a stationary solution, so the profile does not change with time:

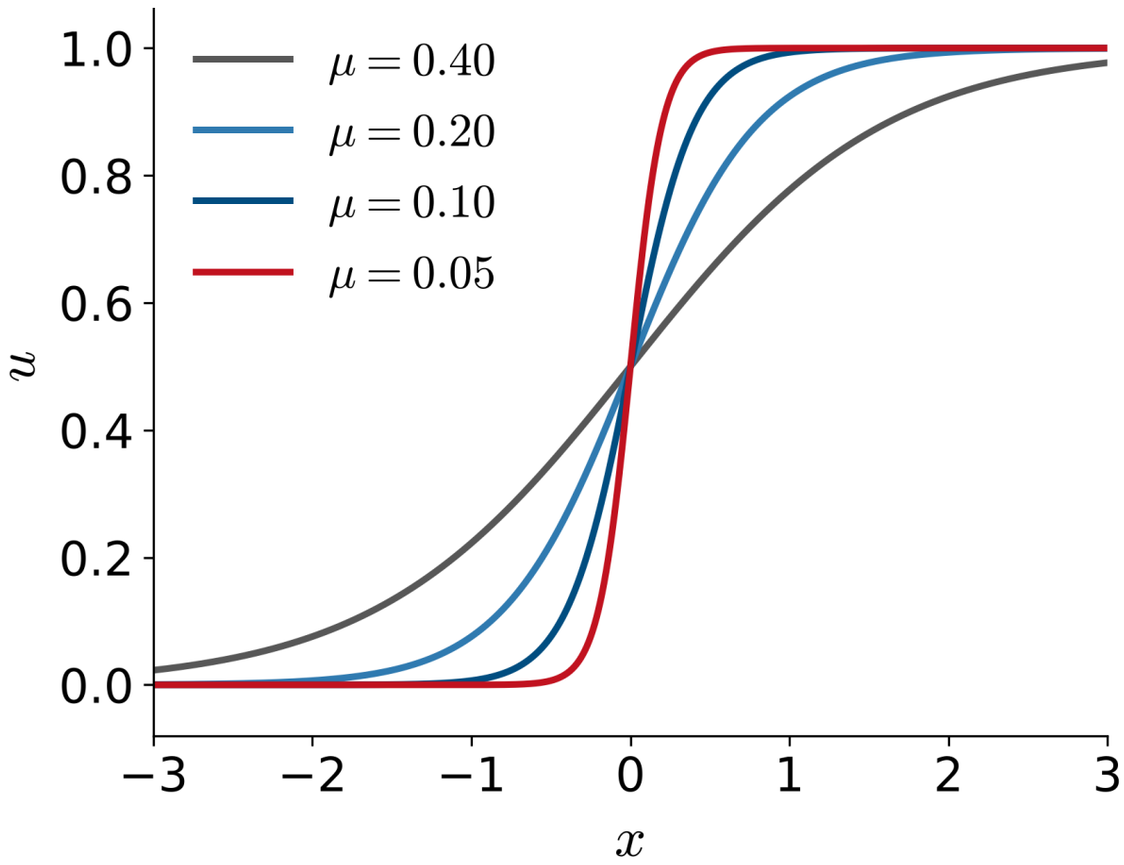
$$c = \tfrac{1}{2}, \quad b = -1 \qquad \Longrightarrow \qquad \boxed{u = -\frac{c}{b}\left[1 + \tanh\frac{c\,(x - x_0)}{2\mu}\right]}$$

- The viscous term spreads the jump into a layer of finite thickness.
- The thickness is proportional to μ.
- The inviscid shock is the limit of this profile as μ goes to zero.

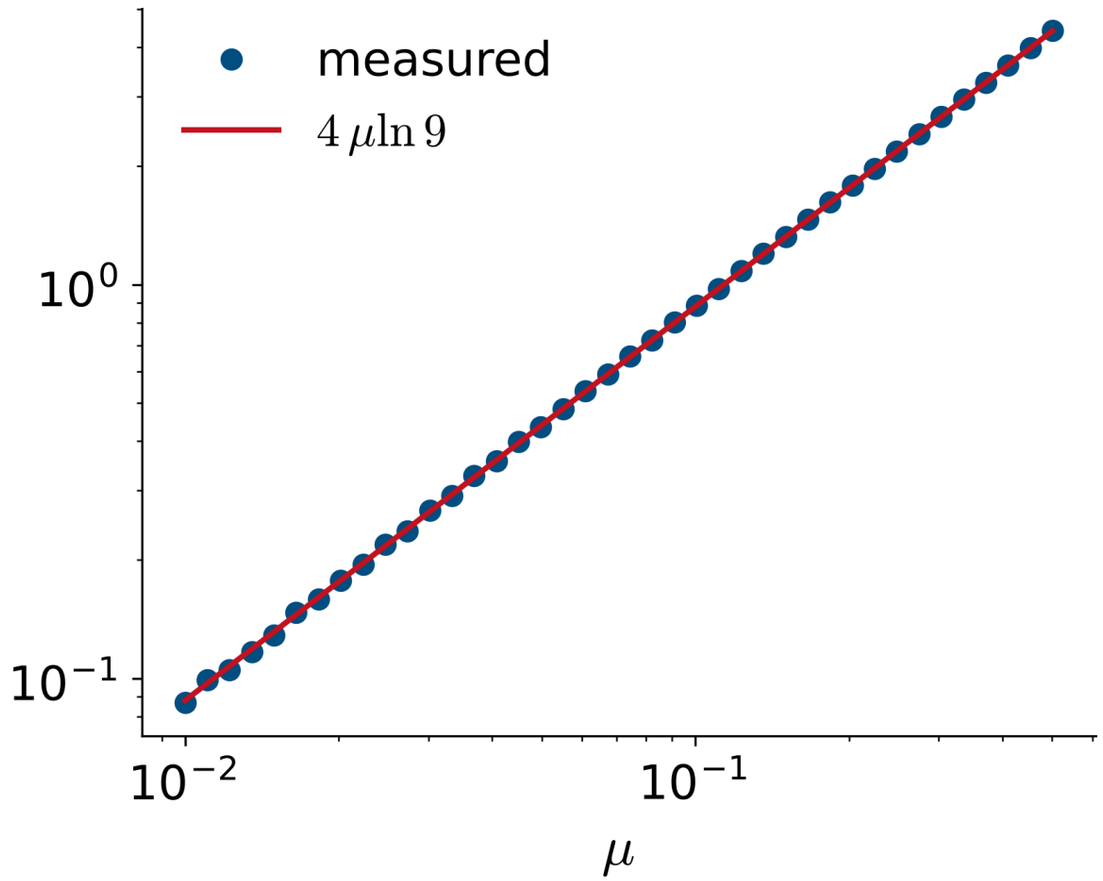
Thickness of the Viscous Layer

The stationary profile for 4 values of μ.

Stationary viscous profile



Shock thickness is set by μ



The shock thickness depends on the dynamic viscosity.

Exact Solutions: Standards for Testing

Steady:

The linearised equation has a closed-form limit as the time grows:

$$u(0,t) = u_0, \qquad u(L,t) = 0$$

$$\implies \quad u = u_0 \, \frac{1 - \exp[R_L(x/L - 1)]}{1 - \exp(-R_L)}, \qquad R_L = \frac{cL}{\mu}$$

Unsteady:

With periodic boundaries and a sine for the initial profile:

$$u(x,0) = \sin kx \qquad \implies \qquad u(x,t) = \exp(-k^2 \mu t) \sin k\,(x - ct)$$

The second is the one we use when measuring temporal accuracy:

The amplitude is known at every time, so the error of a scheme can be separated from the error of the initial profile.

Another solution: Viscous Test with the “saw tooth”

Problem:

The viscous equation on $0 \leq x \leq 2\pi$, periodic, with $\mu = 0.07$.

IC: the Cole-Hopf transformation turns Burgers' equation into the heat equation:

$$u = -2\mu \frac{\partial}{\partial x} \ln \varphi \qquad \Longrightarrow \qquad \frac{\partial \varphi}{\partial t} = \mu \frac{\partial^2 \varphi}{\partial x^2}$$

The choice of φ:

We can use two heat kernels, the second one shifted by 2π for periodicity.

Galilean invariance is also implied, so the saw-tooth is an exact solution at every time, not only at t = 0.

$$\varphi = \exp\left[\frac{-(x-4t)^2}{4\mu(t+1)}\right] + \exp\left[\frac{-(x-4t-2\pi)^2}{4\mu(t+1)}\right]$$

$$u = -\frac{2\mu}{\varphi}\frac{\partial \varphi}{\partial x} + 4$$

Objective:

We wish to check the march against a solution known at every time, and run both forms (conservative and non-conservative) on it.

Reference: Barba’s CFD course: https://github.com/barbagroup/CFDPython

Summary

Burgers' equation carries an unsteady, a nonlinear convective and a diffusive term, and reduces to the equations of Lectures 12 and 13 in two limits.

The **conservation form** and the **non-conservative form** agree wherever u is differentiable and disagree across a jump:

$$\frac{\partial u}{\partial t} + \frac{\partial F}{\partial x} = 0 \qquad \text{and} \qquad \frac{\partial u}{\partial t} + A\,\frac{\partial u}{\partial x} = 0, \qquad A = \frac{dF}{du}$$

A **weak solution** satisfies the integral statement, which contains the flux F.
The jump condition follows from it:

$$s\,[u] = [F] \qquad \Longrightarrow \qquad s = \frac{u_1 + u_2}{2}$$

A scheme in flux-difference form has the conservative property: interior fluxes cancel exactly, and the computed shock takes the **Rankine-Hugoniot** speed.

The non-conservative form has no flux difference to cancel. It converges to the wrong shock speed, and a finer mesh does not fix it.

The viscous term replaces the jump by a layer whose thickness is proportional to μ, and the inviscid shock is its zero-viscosity limit.